

Continual Learning and Forgetting

DATA 443/643 · Week 8, note 8.2

Week 8

American University

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Ideas from Stanford CS224N 2025 L11 (a full Δ is another copy of W) plus the XOR-then-AND lab. Original slides; that course is not copied.

Lecture plan

- 1 Task B walks off task A
- 2 Measure A while you train B
- 3 Replay still moves W
- 4 Freeze W, train an adapter

Key goal: forgetting is A vs B, not train vs test on one task.

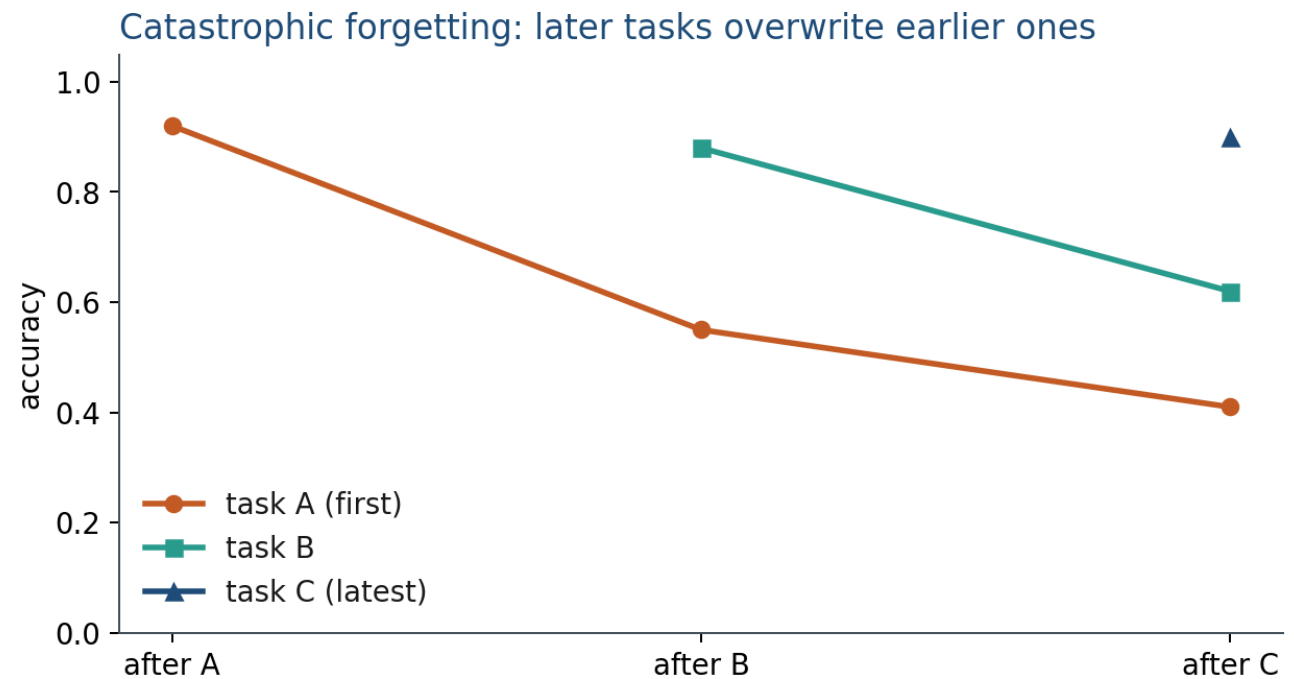
First time: what / why / formula

- What / why. Sequential training without replay overwrites the weights that did A. Continual learning tries to add B without burying A. You will fine-tune. XOR-then-AND in Lab 8 is the cartoon. Production: new policy PDFs every month.
- Architecture. Same net. Mitigations: replay (mix old batches), freeze layers, LoRA (isolate Δ in BA), or a regularizer toward old θ (EWC cartoon: penalty on important weights).
- Formula. Cartoon EWC: $\mathcal{L}_B(\theta) + \frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^A)^2$. You do not compute a full Fisher in this class; you need the idea.
- Tradeoffs. + Replay is simple. – Replay needs stored data; freeze can block B; LoRA still forgets if you merge carelessly; EWC is extra knobs.

How. Measure A after B. If you only plot B, you hid the bug. Replay is the first lever; LoRA is the isolator next note.

A shared W cannot sit at two minima

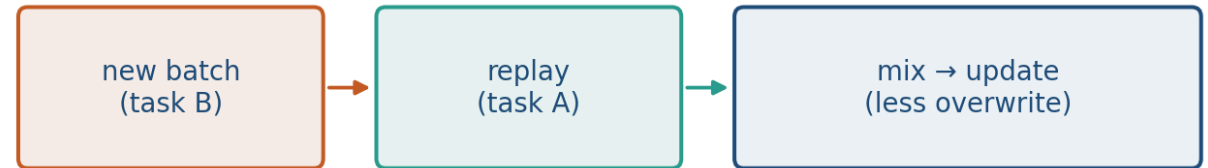
- XOR then AND on the same hidden layer is Lab 8.
- In an LLM the old task is code, languages, refusal style.
- If you only report B, you did not look.



Replay keeps a component of A

- Mix A rows into B batches, or sample the old model.
- 50/50 is the simplest buffer.
- Replay still updates W. That is not LoRA.

Replay (or freeze, or LoRA) is how you keep task A alive



Does W move?

Replay / EWC

Yes. You still need A data (or a Fisher sketch).

Frozen LoRA

No. XOR accuracy on the frozen net cannot move. That is why PEFT exists.

What to take from this lecture

Plot A

Held-out A every k steps of B.

Name it

Overfitting is one task. Forgetting is two.

Lab 8

XOR collapses toward AND unless you replay or freeze.